How Toto-2.0 makes a forecast

Datadog · 22M parameters · decoder-only transformer, time and row attention · zero-shot

1. Targets only, one row each

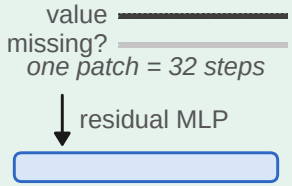
Toto 2.0 reads the targets. Related targets form one group. A ? is a value not known yet.

Covariates are not supported yet; day length and the NAO index are left out.

2. Scale as you go, squash, patch

Each patch is scaled with the mean and standard deviation of the data up to it, then squashed by arcsinh.

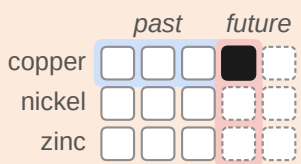
$$\tilde{x} = \operatorname{arcsinh}\left(\frac{x - \mu_{\le k}}{\sigma_{\le k}}\right)$$



one token: 512 numbers

3. Five time blocks, then a row block

The future goes in as masked patches M behind the context. RMSNorm in front, shortcut around.



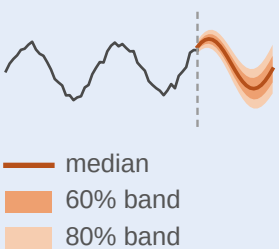
Time attention: causal, along the row (RoPE + xPos)

Row attention: the 6th block, down the column across the group

Feed forward: SwiGLU, 512 → 1,368 → 512

4. Nine quantiles, one pass

Each token gives **9 quantiles** for the 32 steps of the next patch. With the future masked, the whole horizon comes out of **one pass**.



Undoing arcsinh and the scaling restores the units; the median is the point forecast.

A small decoder that reads related targets together and gives the whole horizon in one pass.

6 blocks (5 time, 1 row) · width 512 · 8 heads · patches of 32 · 9 quantiles

In the notebooks: last 2,048 days or full monthly history · five metals together · no covariates · weights: Apache 2.0